

PROBLEM STATEMENT #13 | Healthcare & Telemedicine

Patient Health Record Consent Management System

1. Problem Context & Overview

A patient-centric electronic health data gateway where patients explicitly manage granular, time-bound consent permissions for clinics, diagnostic labs, and consulting doctors to access their medical history.

Target Stakeholders / Actors: *Patient, Clinic Administrator*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall allow patients to grant time-bounded (e.g., 24-hour) access permissions for specific diagnostic records to verified clinic doctors.

Sample Acceptance Criteria: Pass: Clinic access automatically revokes upon expiry timestamp; Fail: Clinic accesses records after consent expiration.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: All consent grant, revoke, and access events shall be permanently written to an append-only audit trail.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.